

Solution Requirements (Functional & Non-functional)

Date	29/09/2026
Team ID	SWTID-2026-7994
Project Name	AI StudyBuddy - AI assistant for smarter learning and study support
Maximum Marks	-

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Register with name, email, password and role (default student); password hashed with bcryptjs
FR-2	User Login and Authentication	Login with email and password; access and refresh JWT tokens returned; protected routes verify the token
FR-3	Role-Based Access Control	Student and Admin roles; admin-only routes return 403 for other users
FR-4	Study Material Management	Upload .txt, .md or .pdf files up to 5 MB; store title and content; retrieve and manage own materials
FR-5	AI Summary Generation	Generate a concise summary of a material through Gemini and save it
FR-6	AI Flashcard Generation	Generate question and answer flashcards from a material
FR-7	AI Quiz Generation	Generate multiple-choice questions with options and correct answers
FR-8	AI Study Plan Generation	Generate a plan from goal, hours per day and number of days
FR-9	History and Saved Resources	View all previously generated resources linked to the user
FR-10	Administration	Admin manages user accounts, monitors AI usage and reviews system logs

Non-functional Requirements:

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple REST endpoints with clear JSON responses and messages; easy to consume from the React frontend and test in Postman.
NFR-2	Security	JWT authentication, role-based authorization, bcryptjs hashing, file type and size validation on uploads, secrets kept in the .env file, CORS restricted to the client URL.
NFR-3	Reliability	Global error handler returns consistent status codes; Mongoose schema validation protects data; server starts only after the database connects.
NFR-4	Performance	Lightweight Express routing; AI calls handled asynchronously (Postman tests showed responses in seconds); uploads limited to 5 MB.
NFR-5	Availability	Stateless server can run as multiple instances behind a load balancer; MongoDB Atlas provides managed availability.
NFR-6	Scalability	Modular controllers, routes, middleware, models and utils allow new AI features to be added with minimal architectural change.